

1 Explainability in Reinforcement Learning

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Chapter 1

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning still face two major limitations. First, there is a lack of intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. Second, methods are proposed generally lack objective metrics for evaluating the quality of the explanations they produce.

To the best of our knowledge, only [Zahavy et al., 2017] have proposed exploiting the latent space learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the latent representations produced by the neural network and organize them into clusters that correspond to high-level concepts. This method falls squarely within the deep clustering approach presented in Section ??.

While the core idea of this paper is innovative and closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representations (CAR) method proposed in this chapter addresses these two limitations. It automates concept discovery by relying on unsupervised clustering algorithms to identify the concept structure of the latent space. It introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policy.

The remainder of this chapter is organized as follows. The first section introduces the theoretical assumptions underlying the proposed CAR method, referred to as the representation convergence hypothesis. The second section describes the CAR method itself, with a detailed account of the extraction of latent representations using surrogate policies, along with the automatic detection of concept structures via clustering. The third section then presents the proposed evaluation metric.

1.1 Representation Convergence Hypothesis

The Clustering Agent Representations (CAR) method relies on a fundamental assumption, referred to as the Representation Convergence Hypothesis. This hypothesis follows the two foundational assumptions established throughout this manuscript.

The first assumption rejects the notion that deep neural networks constitute irreducible black boxes. Adopting a materialist and mechanistic perspective, we consider that every cognitive process results from a sequence of causal computations. Although these computations may be highly complex, they remain explainable through the analysis of the internal mechanisms. Consequently, the opacity of deep neural networks should be understood as an epistemic limitation rather than an ontological one.

The second assumption states that neural networks achieve generalization by learning concepts. During optimization, the network progressively organizes its latent space into representations that capture the information required to solve the task. These concepts are not explicitly specified during training but emerge naturally from the interaction between the learning algorithm.

These two assumptions naturally motivate an intrinsic approach to explainability. If the reasoning process of a neural network is encoded within its latent representations, then explaining the behavior of the agent amounts to understanding the organization of its latent space. This family of approaches is commonly referred to as mechanistic interpretability. We define mechanistic interpretability as the study of the internal organization of latent representations with the objective of identifying the concept structures (Definition) responsible for a model's decisions.

The Representation Convergence Hypothesis extends this mechanistic perspective by introducing an assumption regarding concepts that arise from learning. We hypothesize that, for a given task and a given cognitive process, there exists a privileged conceptual organization that naturally emerges because it constitutes an efficient representation for solving the task. While independently trained neural networks may converge toward different parameter values, they are nevertheless expected to learn conceptually similar representations.

The optimization process is not assumed to converge toward identical weights but toward equivalent organizations of the latent space. Neural network parameters are known to be highly non-identifiable, and multiple parameter configurations may implement the same function. However, then independently trained agents should repeatedly discover comparable conceptual organizations. Representation convergence should be observable at the level of latent concepts, even when the corresponding network parameters differ.

The remainder of this chapter builds upon this hypothesis. In particular, we propose that the degree of convergence between independently learned conceptual organizations provides an objective criterion for assessing the quality of the extracted representations. This principle forms the theoretical foundation of the Clustering Universality metric introduced later in this chapter.

1.2 Method

As discussed in Chapter ??, several hypotheses have been proposed regarding the organization of latent representations in deep neural networks. Following the work of [Zahavy et al., 2017], we therefore analyze the latent space through clustering techniques, placing our approach within the broader field of Deep Clustering (Section ??).

This design choice is motivated by our rejection of the superposition hypothesis and, more generally, of any view that assumes a linearly organized concept structure. We consider this premise unnecessarily restrictive. We only assume that concept representations induce a geometric structure that can be recovered through unsupervised clustering.

The CAR methodology is composed of two main stages. The first stage consists of generating multiple independent neural networks that implement the same decision policy, referred to as surrogate policies. The second stage clusters their latent representations in order to identify the underlying conceptual organization.

1.2.1 Multiple surrogate policies

The core idea of CAR is to evaluate whether independently trained neural networks converge toward similar conceptual organizations. To test this hypothesis, several neural networks must be trained independently and their latent representations compared. If similar conceptual structures consistently emerge, this provides empirical evidence supporting the Representation Convergence Hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. Reinforcement learning, introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. Making it impossible to determine whether differences observed in the latent space originate from the learning dynamics or from genuine differences between the learned policies.

To overcome this limitation, we do not directly compare independently trained reinforcement learning agents. Instead, we first consider a reference policy π that has already been trained. This policy is then imitated using several independently initialized surrogate neural networks trained through supervised behavioral cloning. Since all surrogate models approximate the same target function, they are expected to converge toward equivalent decision policies while preserving independent internal parameterizations.

This protocol provides an experimental setting in which differences between latent representations can be interpreted as resulting from the internal organization learned by each neural network rather than from differences in the underlying policy itself. It becomes possible to investigate whether equivalent policies also exhibit equivalent conceptual organizations.

It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method. In the explainability literature, surrogate approaches replace a complex model with a simpler and inherently interpretable model, such as a decision tree or a linear classifier. This is not

the case here. The surrogate policies remain deep neural networks. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence. The explanations are therefore still extracted directly from the internal representations of deep neural networks, making CAR a fully intrinsic and mechanistic explainability approach.

1.2.2 Training surrogate policies

Throughout the remainder of this manuscript, the initial policy we aim to explain is denoted as π (represented in green in Figure ??). This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [?], alpha-beta algorithm [?], or even from demonstrations provided by humans [Ross and Bagnell, 2010]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π .

The replicated policies are referred to as surrogate policies, denoted by π' (shown in red in Figure ??). For any observation O , the distribution over the action space produced by π' should approximate that of π as closely as possible.

The surrogate policy π' is implemented as a Neural Network (NN) parameterized by Θ . This network is composed of two principal components: an encoder and a decoder. The encoder, represented by the function f_{enc} , maps observations from $\mathcal{O}$ into a Latent Space (LS) $\mathcal{Z}$ (shown in blue in Figure ??). The decoder then reconstructs actions from this latent representation.

To train π' to accurately replicate the action distribution, we define a global loss function $\mathcal{L}$ optimized via gradient descent. This loss, defined in Equation (1.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function p , which maps a distribution to its parameterization. In practice, this function does not need to be explicitly defined, as modern neural policies directly output the parameters of the action distribution. Once training is complete, this loss ensures that π' closely approximates the behavior of the original policy π .

$$\min_{\Theta} \sum_{O \in \mathcal{O}} \mathcal{L}(O) = \min_{\Theta} \sum_{O \in \mathcal{O}} \text{MSE}\left(p(\pi(O)), p(\pi'_{\Theta}(O))\right) \quad (1.1)$$

On the set $\mathcal{E}$, we use a clustering function f_{clust} to partition the point cloud of embedded observations into distinct clusters. Each observation is assigned a cluster label based on its proximity in the LS, resulting in the clustering $\mathcal{C}$. This process produces a structured representation of the groups of observations encountered by the agent, effectively simplifying the environment's representation through the lens of the policy's LS.

To demonstrate this clustering faithfully reflects the agent's internal decision-making process, we introduce a set of evaluation metrics, which will be detailed in the following section.

Bibliography

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